A (

Review - statistics

Assume we have an iid sample $X_1, \dots, X_n$ from a random variable X, where $X \sim f(x)$

→ Use data $X_1, \dots, X_n$ to learn about $f(x)$.

I.- Look at $\bar{X}$, $s^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$.

a).- For any $f(x)$ with mean μ & variance σ^2:

$$E(\bar{X}) = \mu$$
$$Var(\bar{X}) = \sigma^2/n = \frac{Var(X_i)}{n}$$
$$E(s^2) = \sigma^2$$

b).- If $X \sim N(\mu, \sigma^2)$

(i) $\bar{X} \sim N(\mu, \sigma^2/n)$

(ii) $\bar{X}$ independent of s^2

(iii) $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{(n-1)}$

Know:
- If $U \sim N(0,1)$, $V \sim \chi^2_{(p)}$ & U, V independent $\Rightarrow W = \frac{U}{\sqrt{V/p}} \sim t_p$
- " $U \sim \chi^2_{r_1}$, $V \sim \chi^2_{r_2}$ & " $\Rightarrow W = \frac{U/r_1}{V/r_2} \sim F_{r_1, r_2}$
- $U \sim N(0,1) \Rightarrow U^2 \sim \chi^2_1$
- $U_1, \dots, U_m$ iid $N(0,1)$ $\rightarrow \sum_{i=1}^{m} U_i^2 \sim \chi^2_{(m)}$

II.- Now assume we know $f(x|\theta_0)$, but don't know θ_0. E.g., $X \sim N(\mu_0, \sigma_0^2)$, μ_0, σ_0^2 unknown

Goal: Estimate θ_0 using $\hat{\theta}(\underset{\sim}{x})$

① - How to Estimate θ_0 using $X_1, \dots X_n$? $\left\langle \begin{array}{l} \text{MOM}: \hat{\theta}_{MOM} \\ \text{MLE}: \hat{\theta}_{MLE} \end{array} \right.$

a).- MOM.
"Set sample moments = population moments" & solve for θ.

Remember:
- # equations = # parameters
- For example, if 2 parameters:

$$E(x) = \bar{X}$$
$$Var(x) = \tilde{s}^2$$

b).- MLE.

→ Chose value of θ that is more likely to have generated your data.

→ Log-likelihood function: $L(\theta | x_1, \dots, x_n) = \sum_{i=1}^{n} \ln f(x_i; \theta)$

Define $\hat{\theta}_{MLE}$ as:

$$\hat{\theta}_{MLE} = \underset{\theta \in \Theta}{\arg\max}\ L(\theta | x_1, \dots, x_n)$$

(ie., maximize $L(\cdot)$ w.r.t. θ)

* → If $\hat{\theta}_{MLE}$ is the MLE for θ_0 ⇒ MLE of $\gamma_0 = g(\theta_0)$ is $\hat{\gamma}_{MLE} = g(\hat{\theta}_{MLE})$ (called "Invariance Property of the MLE")

→ Score function $S(x,\theta) = \frac{d \ln f(x,\theta)}{d\theta}$. Key: $\mathcal{E}[S(x,\theta_0)] = 0$ (called "Score Identity")

→ Information Matrix Identity:

$$\mathcal{E}[(S(x,\theta_0))^2] = -\mathcal{E}\left[\frac{d^2 \ln f(x,\theta_0)}{d\theta^2}\right]$$

all evaluated at θ_0, the true value of θ

② How to evaluate estimators? (which one is "better"?)

→ $\hat{\theta}$ is unbiased if $\mathcal{E}[\hat{\theta}] = \theta_0$

→ The Mean Square Error is: $MSE(\hat{\theta}) = \mathcal{E}[(\hat{\theta}-\theta_0)^2] = \overbrace{Var(\hat{\theta})}^{\text{precision}} + \overbrace{Bias^2(\hat{\theta})}^{\text{accuracy}}$ } Rule: choose the one with smallest MSE.

↳ Measure of quality of an estimator.

where $Bias(\hat{\theta}) = \mathcal{E}[\hat{\theta}] - \theta_0$.

→ Question: can we find "the best" estimator for θ_0?

Problem: Impossible to find the best estimator (i.e., one with smallest MSE) as compared to any estimator and for any value of θ_0 ⇒ we focus only on unbiased estimators

→ A best unbiased estimator $\hat{\theta}^*$ is one such that:

(i) $\mathcal{E}[\hat{\theta}^*] = \theta_0$

(ii) $Var(\hat{\theta}^*) \leq Var(\hat{\theta})$ for any other unbiased estimator $\hat{\theta}$

$\hat{\theta}^*$ is called UMVUE.

→ Question: How to find a UMVUE?

* → CRLB: Given an iid sample, for any unbiased estimator $\hat{\theta}$ of θ_0:

$$Var(\hat{\theta}) \geq \frac{1}{n\,I(\theta_0)}$$

where $I(\theta_0) = Var(S(x,\theta_0)) = \mathcal{E}[S^2(x,\theta_0)] = -\mathcal{E}\left[\frac{d^2 \ln f(x,\theta_0)}{d\theta^2}\right]$

Hence,

If ① $\mathcal{E}(\hat{\theta}) = \theta_0$ (unbiased)

and ② $Var(\hat{\theta}) = \frac{1}{n I(\theta_0)}$

} ⇒ $\hat{\theta}$ is UMVUE for θ_0, $\hat{\theta} = \hat{\theta}^*$

③ Asymptotic Theory

→ Idea: Sometimes hard to find unbiased estimators or the distribution of $\hat\theta$ ⇒ use approximations as $n\to\infty$ (for large samples)

a).- Where does $\hat\theta$ go?

→ If $\lim_{n\to\infty} \Pr(|X_n-c|<\varepsilon)=1 \Rightarrow X_n \xrightarrow{p} c$.

In words: (the prob. that X_n is arbitrarily close to c goes to 1 as our sample size goes to ∞, i.e., as it gets very large).

→ $\hat\theta$ is consistent for θ_0 if $\hat\theta \xrightarrow{p} \theta_0$.

→ To show consistency:

$$\left.\begin{array}{l}\lim_{n\to\infty} Bias(\hat\theta)=0 \\ \lim_{n\to\infty} Var(\hat\theta)=0\end{array}\right\} \Rightarrow \hat\theta \xrightarrow{p} \theta_0.$$

Remember: $\lim_{n\to\infty} Bias(\hat\theta)=0$ iff $\lim_{n\to\infty} E(\hat\theta)=\theta_0$

★ → If $\hat\theta \xrightarrow{p} \theta_0$ & $g(\cdot)$ is a continuous function then $g(\hat\theta) \xrightarrow{p} g(\theta_0)$

Ex: $\tilde S^2 \xrightarrow{p} \sigma_0^2 \Rightarrow \tilde S \xrightarrow{p} \sigma_0$

→ WLLN: $\bar X \xrightarrow{p} \mu$.

b).- Approximate distribution of $\hat\theta$?

- $X_n \xrightarrow{d} Y$ if $\lim_{n\to\infty} F_{X_n}(x) = F_Y(y)$.
- CLT: $\sqrt{n}(\bar X-\mu) \xrightarrow{d} \underbrace{N(0,\sigma^2)}_{\text{limiting distribution}}$, $\mu=E(X)$, $\sigma^2=Var(X)$

implies

$$\bar X \overset{A}{\sim} \underbrace{N(\mu,\sigma^2/n)}_{\text{asymptotic distribution}}$$

- WLLN and CLT are useful b/c $\hat\theta$ is always a function of sample averages (see future classes)

c).- Other useful Asymptotic results:

(i).- Slutsky's theorem. $X_n \xrightarrow{d} U$, $Y_n \xrightarrow{p} a \Rightarrow Y_n X_n \xrightarrow{d} aU$ $\quad Y_n+X_n \xrightarrow{d} a+U$, where a: constant, U: random variable

★ (ii). Delta Method:

If $\sqrt{n}(X-\mu) \xrightarrow{d} N(0,\sigma^2)$ & $g(\cdot)$ is a function

$$\Rightarrow \sqrt{n}(g(X)-g(\mu)) \xrightarrow{d} N(0,[g'(\mu)]^2\sigma^2)$$

d)- Find $\hat{\theta}$ with minimum asymptotic variance Ω

$$\sqrt{n}(\hat{\theta}-\theta_0) \xrightarrow{d} N(0,\Omega)$$

→ An estimator $\hat{\theta}^*$ is asymptotically efficient IF:

$$\sqrt{n}(\hat{\theta}^*-\theta_0) \xrightarrow{d} N(0,v(\theta_0))$$

$$v(\theta_0) = I^{-1}(\theta_0)$$

i.e.

$$\hat{\theta}^* \overset{A}{\sim} N\Big(\theta_0, \underbrace{\frac{v(\theta_0)}{n}}_{=\text{CRLB}}\Big)$$

→ The MLE $\hat{\theta}_{MLE}$ is:

- consistent
- asymp. normal
- asymp. efficient

$\left.\right\}$ $\sqrt{n}(\hat{\theta}_{MLE}-\theta_0) \xrightarrow{d} N(0, I^{-1}(\theta_0))$.

Note:

· Need to complete "Hypothesis Testing and Confidence Intervals".